ECON 695 Problem Set 3

September 24, 2025

Due: 11:59PM Central Time on Friday, 10/03/2025. Upload a photocopy of your handwritten solutions, or a PDF generated by LaTex/lyx to Canvas.

Note: This problem set will be graded on a complete/incomplete basis. Each question is worth 2 points – you will get full 2 points as long as you attempt to answer the question, 0 if you provide an empty answer.

Note: The questions will help you prepare for the midterm on 10/08. The midterm will have 2-3 questions instead of 5.

1. Suppose you have a random sample $\{(x_i, y_i)\}$ of size N, where each y_i is 1×1 (a scalar outcome), and each x_i is also 1×1 (a scalar regressor). If you run a linear regression of y_i on x_i without a constant, show the regression coefficient (OLS estimator) is:

$$\hat{\beta} = \frac{\frac{1}{N}\sum_i y_i x_i}{\frac{1}{N}\sum_i x_i^2} \tag{1}$$

   Hint: $\hat{\beta} = argmin_b \frac{1}{N}\sum_i (y_i - x_i b)^2$.

2. Prove the Frisch-Waugh theorem for the sample OLS regression coefficients by showing that the j^{th} row of $\hat{\beta}$ is:

$$\hat{\beta}_j = [\frac{1}{N}\sum_{i=1}^{N} \hat{\xi}_i^2]^{-1}[\frac{1}{N}\sum_{i=1}^{N} \hat{\xi}_i y_i] \tag{2}$$

   where $\hat{\xi}_i$ is the *estimated residual* from an auxiliary OLS regression of x_{ji} on all the other $x's$ (denoted by $x_{(-j)i}$):

$$x_{ji} = x'_{(\sim j)i}\hat{\pi} + \hat{\xi}_i. \tag{3}$$

   HINT: review lecture 4.

3. We observe an outcome variable y_i and covariates x_i and z_i in a sample of N observations. Assume that the sample means of x_i and z_i are both 0 (i.e., $\overline{x} = 0$, $\overline{z} = 0$). Consider the regression model:

$$y_i = a_0 + a_1 x_i + v_i \tag{4}$$

   which we estimate by OLS, obtaining OLS regression coefficients $\widehat{a}_0$ and $\widehat{a}_1$. Show that if x_i is *orthogonal* to z_i in the observed sample ($\frac{1}{N}\sum_i x_i z_i = 0$), then when you estimate the regression model:

$$y_i = b_0 + b_1 x_i + b_2 z_i + e_i \tag{5}$$

   by OLS, you will get

$$\widehat{b}_0 = \widehat{a}_0, \text{ and } \widehat{b}_1 = \widehat{a}_1$$

   HINT: Write out the three FOC's for the OLS coefficients from model (5). Compare these with the two FOC's from model (4) and use the information you are given.

4. Suppose you can observe the wages, education (yrs of schooling), immigrant status, and race and ethnicity of a random sample of workers.

   (a) Consider two linear regressions of $y_i = \text{log wage}_i$ on observable characteristics:

   $$y_i = \lambda_0 + \lambda_1 Imm_i + u_i \text{ (short model)}$$
   $$y_i = \beta_0 + \beta_1 Imm_i + \beta_2 Educ_i + \epsilon_i \text{ (long model)}$$

   If you estimate the regressions in the random sample, how does the OLS estimator $\hat{\lambda}_1$relate to $\hat{\beta}_1$? Which assumption do you need to have $\hat{\lambda}_1 > \hat{\beta}_1$?
   Hint: use the omitted variable bias, and define the terms in the formula carefully.

   (b) Split the immigrants into 3 mutually exclusive groups:

      i. Asian immigrants: workers who answer asian = 1, hispanic =0, immigrant =1;
      ii. Hispanic immigrants: workers who answer hispanic=1, immigrant=1;
      iii. Other immigrants: workers who answer asian=0, hispanic=0, immigrant=1.

   Specify a single regression that allows you to estimate the mean log wage of each immigrant group relative to natives (immigrant=0).

   (c) How much of the (log) wage gaps between each immigrant group and native workers can be explained by differences in education attainment? Specify the regression(s) you will need to estimate to answer this question, and provide an answer from your regressions.
   Hint: consider a pooled regression with education as a control, or separate regressions by immigrant group. Focus on the "between" component of the decomposition.

5. You are tasked to provide a causal analysis of how the inflow of Asian immigrants affects the earnings of native workers. Assume you have a county c-level data $\{y_c, x_c, z_c\}$:

   - y_c: the change in real log wage among native workers in county c from 1990 to 2000;
   - x_c: the percent change in the number of immigrants in county c from 1990 to 2000 (if N_{ct} denote the number of immigrants in county c in year t, $x_c = 100 * (N_{c,2000} - N_{c,1990})/N_{c,1990}$.
   - z_c: the share of restaurants in county c that are owned by Asian immigrants in 1990.

   The causal model of interest is:

   $$y_c = \beta_1 + \beta_2 x_c + u_c \tag{6}$$

   in which β_2 is the causal effect of a 1% increase in immigrant population on the change in log wage of native workers, holding all else constant.

   (a) Consider an OLS regression of y_c on x_c,

   $$\hat{y}_c = \hat{\beta}_1 + \hat{\beta}_2 x_c \tag{7}$$

   how would you interpret the coefficient $\hat{\beta}_2$ on x_c? If $cov(x_c, u_c) = 0$, show that $\hat{\beta}_2$ is a consistent estimator for β_2 in (6).

   (b) If $E[u_c x_c] \neq 0$ in model (6), you may want to look for an instrument for the endogenous x_c. What assumptions do you need for z_c as defined to be a valid instrument for x_c?

   (c) Suppose the assumptions for z_c to be an valid instrument for $x_c$are satisfied. Consider the first-stage regression in the population

   $$x_c = \pi_0 + \pi_1 x_c + \nu_i$$

   and the reduced-form regression (population)

   $$y_c = \delta_0 + \delta_1 z_c + \epsilon_i.$$

   Show:

   $$\frac{\delta_1}{\pi_1} = \beta_2.$$

(d) Do you think z_c as defined is a good instrument for x_c? Why or why not? Hint: this is an open-ended question. You can argue either way.